

The Chain Rule

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Recall that in calculus I, we used the chain rule to find the derivatives of composite functions.

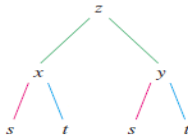
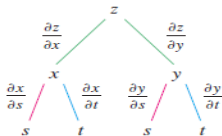
The Chain Rule: Case I Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t . Moreover

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

The Chain Rule: Case II Suppose that $z = f(x, y)$, $x = g(s, t)$ and $y = h(s, t)$ where all functions f, g, h are differentiable. Then z is a differentiable function of s and t . Moreover

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} \quad \text{and} \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

For the chain rule, we can make use of the **tree notation**.

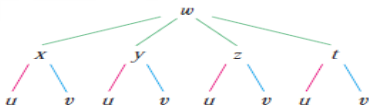


Example 1. Use the chain rule to find $\frac{dz}{dt}$ or $\frac{dw}{dt}$ if

- $z = \sqrt{x^2 + y^2}$, $x = e^{2t}$, $y = e^{-2t}$
- $w = xy + yz^2$, $x = e^t$, $y = e^t \sin(t)$, $z = e^t \cos(t)$
- $z = \sec(x + y) \sin(x - y^2)$, $x = \pi t$, $y = \sqrt{t}$

Note that the chain rule can be generalized to functions with more variables.

- $\models w = f(x, y, z, t), \quad x = x(u, v), \quad y = y(u, v), \quad t = t(u, v)$



- $\models u = u(x, y, z), \quad x = x(r, s, t), \quad y = y(r, s, t), \quad z = z(r, s, t)$



Example 2 Use the chain rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ if

○ $z = \frac{x}{y}$, $x = se^t$, $y = 1 + se^{-t}$

○ $z = e^{xy} \tan(y)$, $x = s + 2t$, $y = \frac{s}{t}$

Example 3. If $z = f(x, y)$ is differentiable with $x = g(t)$, $y = h(t)$ with $g(3) = 2$, $g'(3) = 5$, $h(3) = 7$, $h'(3) = -4$, $f_x(2, 7) = 6$, $f_y(2, 7) = -8$.

Find $\frac{dz}{dt}$ when $t = 3$.

Example 4. If $W = F(u(s, t), v(s, t))$ is differentiable with
 $u(1, 0) = 2$, $u_s(1, 0) = -2$, $u_t(1, 0) = 6$, $v(1, 0) = 3$, $v_s(1, 0) = 5$, $v_t(1, 0) =$
 4 , $F_u(2, 3) = -1$, $F_v(2, 3) = 10$, find $W_s(1, 0)$ and $W_t(1, 0)$.

Example 5. Assume that all the given functions below are differentiable. Use a tree diagram to write the chain rule for the given case.

- $z = f(x, y), \quad x = x(r, s, t), \quad y = y(r, s, t)$
- $w = f(x, y, z), \quad x = x(t, u), \quad y = y(t, u), \quad z = z(t, u)$

Example 6. Use the chain rule to find the given partial derivatives.

- If $u = \sqrt{r^2 + s^2}$, $r = y + x \cos(t)$, $s = x + y \sin(t)$, find $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial u}{\partial t}$ when $x = 1$, $y = 2$, $t = 0$.
- If $M = xe^{y-z^2}$, $x = 2uv$, $y = u - v$, $z = u + v$, find $\frac{\partial M}{\partial u}$, $\frac{\partial M}{\partial v}$ when $u = 3$, $v = -1$.

Implicit Differentiation

Proposition Consider an implicit function $F(x, y) = 0$.
Then we have

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}$$

Proof

Example 1. Find $\frac{dy}{dx}$ if

- $y^5 + x^2y^3 = 1 + ye^{x^2}$
- $\sin(x) + \cos(y) = \sin(x) \cos(y)$
- $\cos(x - y) = xe^y$

Implicit Differentiation

Proposition Consider an implicit function $F(x, y, z) = 0$.
Then we have

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$$

Example 8. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if

- $xyz = \cos(x + y + z)$
- $\sin(x) + \cos(z) = \sin(x) \cos(y)$
- $yz = \ln(x + z)$

Example 9. If $z = f(x - y)$, then show that $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$.

Further Problems

Partial derivatives are used mostly in the solutions of PDEs.

Example 10. If $u = f(x, y)$ where $x = e^s \cos(t)$ and $y = e^s \sin(t)$, show that

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2 \right]$$

Example II. Show that $z = f(x + at) + g(x - at)$ satisfies

$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

where a is a constant.

Hint. Make the change of variables

$$u = x + at \quad \text{and} \quad v = x - at$$

Example 12. Let $z = f(x, y)$ where $x = s + t$ and $y = s - t$. Show that

$$\left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2 = \frac{\partial z}{\partial s} \frac{\partial z}{\partial t}$$

Example 13. Let $z = f(x, y)$ where $x = g(s, t)$ and $y = h(s, t)$. Find

$$\frac{\partial^2 z}{\partial t^2}, \quad \frac{\partial^2 z}{\partial s \partial t}, \quad \frac{\partial^2 z}{\partial s^2}$$

Example 14. Suppose that f and g are two differentiable functions. If the function

$$F := xf(x+y) + yg(x+y)$$

satisfies

$$\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} - \alpha^2 \frac{\partial^2 F}{\partial x \partial y} = 0$$

find value(s) of α .

Example 15. Suppose that f is a twice differentiable function, and $\alpha \in \mathbb{R}$.

Let $u = f(x + \alpha t)$. If

$$\frac{\partial^2 u}{\partial t^2} - 9 \frac{\partial^2 u}{\partial x^2} = 0$$

, then find the value(s) of α .

Example 16. If $f(x, y)$ is harmonic, show that $f(x^2 - y^2, 2xy)$ is also harmonic.

Solution. Let $u = x^2 - y^2$ and $v = 2xy$. If $z = f(u, v)$, then

$$\frac{\partial z}{\partial x} = 2xf_u(u, v) + 2yf_v(u, v) \quad \text{and} \quad \frac{\partial z}{\partial y} = -2yf_u(u, v) + 2xf_v(u, v)$$

$$\begin{aligned}\frac{\partial^2 z}{\partial x^2} &= 2f_u(u, v) + 2x(2xf_{uu}(u, v) + 2yf_{uv}(u, v)) \\ &\quad + 2y(2xf_{vu}(u, v) + 2yf_{vv}(u, v)) \\ &= 2f_u(u, v) + 4x^2 f_{uu}(u, v) + 8xyf_{uv}(u, v) + 4y^2 f_{vv}(u, v),\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial^2 z}{\partial y^2} &= -2f_u(u, v) - 2y(-2yf_{uu}(u, v) + 2xf_{uv}(u, v)) \\ &\quad + 2x(-2yf_{vu}(u, v) + 2xf_{vv}(u, v)) \\ &= -2f_u(u, v) + 4y^2 f_{uu}(u, v) - 8xyf_{uv}(u, v) + 4x^2 f_{vv}(u, v).\end{aligned}$$

Therefore,

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 4(x^2 + y^2)(f_{uu}(u, v) + f_{vv}(u, v)) = 0$$

because f is harmonic. Thus, $z = f(x^2 - y^2, 2xy)$ is a harmonic function of x and y .

Example 17. If $z = f(x, y)$ has continuous partial derivatives of second order, and if $x = r \cos(\theta)$ and $y = r \sin(\theta)$, then find $\frac{\partial^2 z}{\partial r^2}$ and $\frac{\partial^2 z}{\partial \theta^2}$

Solution. Note that, $x = r \cos(\theta)$ and $y = r \sin(\theta)$ imply that

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial x}{\partial \theta} = -r \sin \theta, \quad \frac{\partial y}{\partial r} = \sin \theta, \quad \frac{\partial y}{\partial \theta} = r \cos \theta.$$

Thus,

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial z}{\partial y}.$$

$$\begin{aligned} \frac{\partial^2 z}{\partial r^2} &= \cos \theta \frac{\partial}{\partial r} \frac{\partial z}{\partial x} + \sin \theta \frac{\partial}{\partial r} \frac{\partial z}{\partial y} \\ &= \cos \theta \left(\cos \theta \frac{\partial^2 z}{\partial x^2} + \sin \theta \frac{\partial^2 z}{\partial x \partial y} \right) + \sin \theta \left(\cos \theta \frac{\partial^2 z}{\partial x \partial y} + \sin \theta \frac{\partial^2 z}{\partial y^2} \right) \\ &= \cos^2 \theta \frac{\partial^2 z}{\partial x^2} + 2 \cos \theta \sin \theta \frac{\partial^2 z}{\partial x \partial y} + \sin^2 \theta \frac{\partial^2 z}{\partial y^2}. \end{aligned}$$

Example 17. Cont'd Similarly,

$$\frac{\partial z}{\partial \theta} = -r \sin \theta \frac{\partial z}{\partial x} + r \cos \theta \frac{\partial z}{\partial y}.$$

and

$$\begin{aligned} \frac{\partial^2 z}{\partial \theta^2} &= -r \left(\cos \theta \frac{\partial z}{\partial x} + \sin \theta \frac{\partial}{\partial \theta} \frac{\partial z}{\partial x} \right) + r \left(-\sin \theta \frac{\partial z}{\partial y} + \cos \theta \frac{\partial}{\partial \theta} \frac{\partial z}{\partial y} \right) \\ &= -r \frac{\partial z}{\partial r} - r \sin \theta \left(-r \sin \theta \frac{\partial^2 z}{\partial x^2} + r \cos \theta \frac{\partial z}{\partial y \partial x} \right) \\ &\quad + r \cos \theta \left(-r \sin \theta \frac{\partial z}{\partial x \partial y} + r \cos \theta \frac{\partial^2 z}{\partial y^2} \right) \\ &= -r \frac{\partial z}{\partial r} + r^2 \left(\sin^2 \theta \frac{\partial^2 z}{\partial x^2} - 2 \sin \theta \cos \theta \frac{\partial z}{\partial x \partial y} + \cos^2 \theta \frac{\partial^2 z}{\partial y^2} \right). \end{aligned}$$